Projects - Financial econometrics

Topic 1. Test the effect of pandemic on the volatility in different countries/sectors
The aim is to investigate the effect of the COVID pandemic on the financial market in terms of volatility in different countries/sectors.
For the data set, you may choose the stock market indices from two countries/sectors of your choice for the period of 2015-2024.
You are expected to do a statistical test of the hypothesis of the equality of the volatility between two time series. Specifically, you will test (i) whether there is any difference in the volatility between 2015-2019 and 2020-2024 and (ii) whether there is any difference in the volatility between two countries in the same period.
You will be able to formulate the hypotheses in terms of the equality of the means of correlated data. It should be taken into account that the squared returns exhibit correlations, and thus the usual statistical test for equality between the means of two i.i.d. sequences should be modified accordingly. In particular, you will compute the p-value of the test result.

Topic 2. Volatility and risk management for daily time series (choose one)

1. **Estimation of volatility and Risk management.** We have discussed several approaches to estimating volatility and risk management, namely
   — Parametric method
   — Nonparametric method
   — Extreme value theory
   — Time series models
   The purpose is to compare these approaches for the purpose of risk management.
   (a) Give a summary of each method, by properly formulating the underlying modelling assumptions and potential model diagnostics.
   (b) For the purpose of comparison, we concentration on the last two methods. Implement these methods.
   (c) Evaluate the performance of volatility forecasting to your data and summarize your findings.
2. **Variation of GARCH models and the leverage effect** The aim is to investigate whether an extension of GARCH-model is needed to account for asymmetric effect of the negative returns. The Threshold-GARCH (1,1) model is defined as

$$\begin{aligned}\epsilon_t &= \sigma_t Z_t, \qquad (Z_t)_t \text{ i.i.d. } \mathcal{N}(0,1)\\ \sigma_t^2 &= \omega + \alpha_- \epsilon_{t-1}^2 1_{\epsilon_{t-1}<0} + \alpha_+ \epsilon_{t-1}^2 1_{\epsilon_{t-1}>0} + \beta\sigma_{t-1}^2.\end{aligned}$$

   You are expected to develop codes to fit the T-GARCH model, using the (pseudo) log-likelihood method. And you will test the hypothesis :

$$H_0 : \alpha_- = \alpha_+$$

   You may also consider the E-GARCH model introduced in the class as an alternative.

Topic 3. High-frequency time series modelling (choose one)

1. **Intraday volatility estimation and assessment of microstructure noise** The aim is to estimate the daily volatility from high frequency time series data and to assess the effect of the microstructure noise. Investigate whether there is a need to account for micro-structure noise and demonstrate how to improve your estimation. Based on your analysis, suggest a method to compute VaR with high frequency time series data.
2. **GARCH models and Stochastic Volatility models for high frequency returns** We have shown the link between stochastic volatility models and GARCH models. The aim is to assess the utility of the GARCH model for high frequency return series.

   To properly study the theoretical link we first use simulated data. You will simulate stochastic volatility models in a high frequency and use the subsampled series to fit the GARCH model successively. Study the convergence of the parameters as a function of sampling frequency. Is the result of your experiment coherent with the theory ? Repeat the experiment to quantify the variability of the parameter estimates.

   Once you have obtained a good understanding of what is expected, apply the same procedure to real data. What can you conclude ? Based on your analysis, suggest a method to compute VaR with high frequency time series data.
3. **Functional data analysis for intraday cumulative log returns** Under the functional data framework, the data can be expressed as

$$Y_n(t_{nj}) = X_n(t_{nj}) + \varepsilon_{nj}\,, \quad n = 1, \ldots, N\,, j = 1, \ldots, J_n\,,$$

   where $\varepsilon_{nj} \sim (0, \sigma^2)$ and

$$X_n(t) = \mu(t) + \sum_{k=1}^{p} \xi_{nk}\psi_k(t)\,.$$

   Give an interpretation of the eigenfunctions and assess if ξ_{nk} follows a normal distribution for each k. Also examine if $(\xi_{nk})_{n=1,\ldots,N}$ can be considered as a white noise process. Based on your analysis, give an estimate of the noise variance and assess the quality of the fit as a function of p. Now, consider a time-series model for the sequence of the coefficients $(\xi_{n,k})_{n=1,\ldots,N}$ to develop forecasting methods :

$$\widetilde{X}_{N+h|N}(t) = \mu(t) + \sum_{k=1}^{p} \tilde{\xi}_{N+h,k}\psi_k(t)\,,$$

   where $\tilde{\xi}_{N+h,k}$ is h-step ahead forecasts from $\xi_{N,k}$. Evaluate prediction performance. You may use the first few years of data to build your model and the later year to evaluate prediction performance.

Topic open : Generative models You are asked to generate realistic financial data needed for the development of novel financial products. Given what you have learned or done, provide your strategy to help your colleagues. Generate your own data and give your reasoning to do so.

Remark : Wherever real data is used, you need to include exploratory data analysis, by clarifying what information you can extract from your data and giving an informative summary using appropriate numerical and graphical methods.